

Chamber Data Sheet RUN TYPE:

A=Nitrogen Fixation

B=Denitrification

PG _____ of _____

Site: Kings Creek (High), KSDate: 13 Jun 2018Personnel: Neuvoud, Elberhard

Chamber r	Sample Type	MIMS							GC					
		Initial BP	Initial Time	Final time	Final BP	Final Temp	Exetainer 1	Exetainer 2	Exetainer 3	Time of initial sample	Initial Serum Vial	Final Temp (C)	Time of final sample*	Final Serum Vial
D1	BNF	734.1	8:39	10:41	734.2	23.4	20510	10486	10177	10:42	1397	23.8	12:42	164
D2	BNF	734.1	8:41	10:43	734.2	23.5	10672	10997	10380	10:44	1623	23.6	12:44	1500
D3	BDU	734.1	8:42	10:45	734.3	23.6	30304	10150	10369	10:47	678	23.8	12:47	780
D4	inlet	734.1	8:42	8:50	734.1	23.3	10749	10356	30190					
D5	NF	734.1	8:43	10:47	734.3	23.6	10069	30151	10154	10:50	1412	23.8	12:50	1609
D6	BNF	734.1	8:44	10:50	734.3	23.6	10975	30343	10381	10:53	1910	23.8	12:53	1388
D7	inlet	733.9	8:46	9:00	734.1	23.2	20124	10118	20361					
D8	BNF	733.9	8:46	10:52	734.4	23.6	10149	30841	20445	10:55	1431	23.8	12:55	1147
D9	DA	734.0	8:48	10:55	734.2	23.6	30558	10733	30596	10:56	1542	23.8	12:58	756
D10	DU	734.0	8:48	10:58	734.2	23.7	10693	30502	31097	11:01	278	23.8	13:01	1368
D11	DU	734.1	8:49	11:00	734.2	23.7	20011	10980	30228	11:05	1169	23.8	13:05	1594
D12	NF	734.1	8:50	11:03	734.3	23.7	10584	10539	31125	11:07	1454	23.9	13:07	1766
B1	PA	734.2	8:50	11:05	734.4	23.5	10074	10506	20143	11:10	1491	23.9	13:10	1240
B2	BPA	734.2	8:51	11:07	734.1	23.5	10139	10921	31065	11:12	1729	23.9	13:12	1961
B3	NF	734.1	9:02	11:09	734.1	23.8	20375	30567	31090	11:14	1347	23.9	13:14	1127
B4	NF	734.1	9:02	11:11	734.1	23.8	20900	20533	30128	11:16	1434	23.9	13:16	276
B5	DU	734.1	9:03	11:13	734.1	23.7	20516	20148	30178	11:18	893	24.0	13:18	1571
B6	inlet	734.1	9:04	11:15	734.2	23.7	10761	20355	31055	11:20	1460	24.0	13:20	176
B7	inlet	734.2	9:04	11:18	734.2	23.7	20519	20946	10648	11:21	1363	24.0	13:21	1496
B8	inlet	734.1	9:05	9:12	734.2	23.0	30523	31053	30569					
B9	DU	734.1	9:06	11:20	734.3	23.7	20096	10773	30474	11:22	846	24.0	13:22	1757
B10	DA	734.1	9:06	11:22	734.3	23.7	30834	30537	10402	11:24	930	24.0	13:24	313
B11	DA	734.1	9:07	11:24	734.3	23.6	10970	30562	10038	11:26	625	23.8	13:26	1043
B12	inlet	734.1	9:08	9:14	734.2	22.8	10282	31126	30335					

Rock

1
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Chamber Data Sheet RUN TYPE:

A=Nitrogen Fixation

B=Denitrification

PG _____ of _____

Site: Kings Creek (High) KSDate: 13 Jun 2018Personnel: Navorski, Eberhard

MIMS										GC				
Chamber	Sample Type	Initial BP	Initial Time	Final time	Final BP	Final Temp	Exetainer 1	Exetainer 2	Exetainer 3	Time of initial sample	Initial Serum Vial	Final Temp (C)	Time of final sample	Final Serum Vial
11	C1 BNF	734.2	09:11	11:26	734.4	23.5	10809	20751	30816	11:27	1569	24.0	13:27	1929
12	C2 BDA	734.2	09:11	11:29	734.4	23.6	20389	30527	30522	11:31	1450	24.0	13:31	1871
13	C3 initial	734.2	09:13	9:24	734.3	23.3	30535	10947	10242					
14	C4 NF	734.2	9:14	11:31	734.5	23.6	31108	30453	10716	11:32	811	24.0	13:32	273
15	C5 DA	734.1	9:12	11:33	734.5	23.6	50302	31101	10437	11:34	198	24.1	13:34	1388
16	C6 BDA	734.2	9:14	11:35	734.5	23.6	30873	30354	30556	11:37	1433	24.0	13:37	859
17	C7 BDA	734.2	9:15	11:36	734.5	23.6	10929	30341	30424	11:39	1576	24.1	13:39	1082
18	C8 initial	734.2	9:15	9:26	734.3	22.9	30565	20286	30199	11:41	1445	24.1	13:41	1573
19	C9 DU	734.2	9:16	11:39	734.5	23.7	10420	10135	20343	11:42	1628	24.1	13:41	1573
20	C10 DU	734.2	9:17	11:41	734.5	23.8	20122	300303	20413	11:44	1848	24.6	13:43	571
21	C12 NF	734.2	9:17	11:43	734.6	23.7	20334	30573	10493	11:47	1210	24.4	13:45	1576
22	C16 initial	734.2	9:18	11:45	734.6	23.7	10617	20600A	30103	11:49	1566	24.2	13:47	861
23	A1 DA	734.3	9:19	11:47	734.6	23.8	20595	20543	31121	11:49	1313	24.2	13:49	1574
24	S2 NF	734.3	9:19	11:49	734.6	23.8	31106	20360	10036	11:51	1516	24.1	13:50	1172
25	D3 BNF	734.3	9:20	11:50	734.6	23.7	20479	30318	31003		1516	24.1	13:51	1446
26	A4 DU	734.3	9:21	9:28	734.3	23.0	30508	10872	30874	11:54	1352	24.1	13:54	1867
27	A5 initial	734.3	9:21	11:52	734.5	23.8	30506	30219	30770					
28	A6	734.3	9:22	11:54	734.5	23.8	10708	31072	1330	11:55	1401	24.0	13:55	DM009
	A7													
	A8													
	A9													
	A10													
	A11													
	A12													

-19 broke before microbe sample

Chamber Data Sheet		RUN TYPE:		A=Nitrogen Fixation		B=Denitrification		PG _____ of _____	
Site:		1401501 KS Kings KS							
Date:		13 Jun 2018							
Personnel:		Neuraski Eberhard							

Chamber	Sample Type	Sed Height 1 (cm)	Sed Height 2 (cm)	Sed Height 3 (cm)	Sed Height 4 (cm)	Water Volume (ml)	Chamber	Air Volume (mL)	Water Volume (mL)	Chlor 1 Volume (mL)	Chlor 2 Volume (mL)	Rock Volume (mL)	
11 A1	48					4400	Q1 NF	248	360			60	
12 A2	48					430	Q2 NF	248	380			40	
13 A3		8.0	7.8	7.4	7.5		C3	248	1840			80	
14 A4	48	7.3	7.4	7.8	7.4		C4 air	248	360			40	
15 A5	48						C5 air	248	370			30	
16 A6	48	7.6	7.5	7.6	7.4		C6		360			30	Canopy cover
17 A7	48					430	C7	248	360			60	N
18 A8		7.6	7.8	7.8	8.1		C8	248	400			40	E
19 A9	48	5.5	6.0	6.4	6.3	BROKE	C9	248	410			20	S
20 A10	48	5.5	5.0	5.1	5.0		C10		320			50	W
21 A11	48	3.9	3.8	3.1	3.2		C12						Discharge
22 A12	48	5.4	5.6	5.2	5.4		C16						width
23 B1	48	3.0	2.7	2.8	2.8		D1 B	200	1650			400	dist
24 B2	48	8.7	8.1	8.5	8.5		D2 air	200	1650	50	50	340	depth
25 B3	48					430	D3 B	200	1670			280	vel
26 B4	48	4.3	4.5	4.7	4.8		D4		1790	50	50	240	
27 B5		4.0	3.9	3.6	3.6		P5 NF	200	1790	50	50	300	
28 B6	air 48	2.4	2.6	2.2	2.1		D6 air	200	1850	50	50	80	
B7							D7		1880	50	50	200	
B8							D8 B	200	1890			200	
B9							D9	200	1580	50	50	340	
B10							D10	200	1640	50	50	400	
B11							D11	200	1550	50	50	400	
B12							D12 NF	200	1530	50	50	340	
							FI	200	1520	50	50	420	

Chamber Data Sheet		RUN TYPE:		A=Nitrogen Fixation		B=Denitrification		PG ____ of ____	
Site:		Kings, KS							
Date:		13 Jun 2018							
Personnel:		Nevorski, Eberhard							
								lat	
								lon	
A1	B1	C1	D1	Canopy Cover (%): TimeLight Intensity					
A2	B2	C2	D2						
A3	B3	C3	D3						
A4	B4	C4	D4	NH4					
A5	B5	C5	D5	SRP					
A6	B6	C6	D6	TDP					
A7	B7	C7	D7	NO3					
A8	B8	C8	D8	DOC					
A9	B9	C9	D9	Discharge					
A10	B10	C10	D10	Stram Width (cm):					
A11	B11	C12	D11	PointDistance (cm)Ceptth (cm)Flow					
A12	B12	C16	D12	1					
				2					
				3					
				4					
				5					
				6					
				7					
				8					
				9					
				10					

Kings Creek - KS High N 6-11-18

GPS: 39.10658°N
96.60629°W

0 width = 1.70 m

0.2m	0.4m	0.6m	0.8m	1.0m	1.2m	1.4m
10cm	16cm	16cm	16cm	10cm	6cm	3cm

canopy cover:
91, 89, 60, 69

1 width = 3.50 m shallow much, rock

0.5m	1.0m	1.5m	2.0m	2.5m	3.0m
26cm	45cm	50cm	45cm	30cm	14cm

GPS: 39.10653°N
96.60632°W

canopy cover: 94, 91, 84, 92

2 width = 4.70 m

0.5m	1.0m	1.5m	2.0m	2.5m	3.0m	3.5m	4.0m	4.5m
30cm	52cm	53cm	47cm	42cm	38cm	28cm	20cm	7cm

GPS: 39.10645°N
96.60636°W

canopy cover: 96, 94, 80, 6

3 width = 4.1 m

0.5m	1.0m	1.5m	2.0m	2.5m	3.0m	3.5m	4.0m
22cm	32cm	32cm	30cm	23cm	18cm	13cm	3.0cm

GPS: 39.10645°N
96.60644°W

canopy cover: 89, 96, 84, 4

4 width = 6.0 m bubble sed

0.5m	1.0m	1.5m	2.0m	2.5m	3.0m	3.5m	4.0m	4.5m	5.0m	5.5m
29cm	35cm	36cm	34cm	37cm	40cm	42cm	42cm	37cm	31cm	16cm

GPS: 39.10643°N, 96.60645°W

canopy cover = 89, 93, 86, 66

5 width = 1.00 m algae sed

0.2m	0.4m	0.6m	0.8m	1.0m
8cm	8cm	6cm	4cm	

GPS: 39.10641°N, 96.60643°W
canopy cover: 81, 95, 72, 40

6 width = 1.30 m

0.2m	0.4m	0.6m	0.8m	1.0m	1.2m
29cm	10cm	16cm	11cm	8cm	5cm

2.5 more water till end
GPS: 39.10637°N, 96.60645°W
canopy cover: 61, 93, 70, 93

37.5m long, started at 5m rocky, stirs

Filtered: 1014, 1163, 59, 7584, 1353

Unfiltered: 7835, 1476

Kings Creek - KS High N 6-11-18 Eberhard

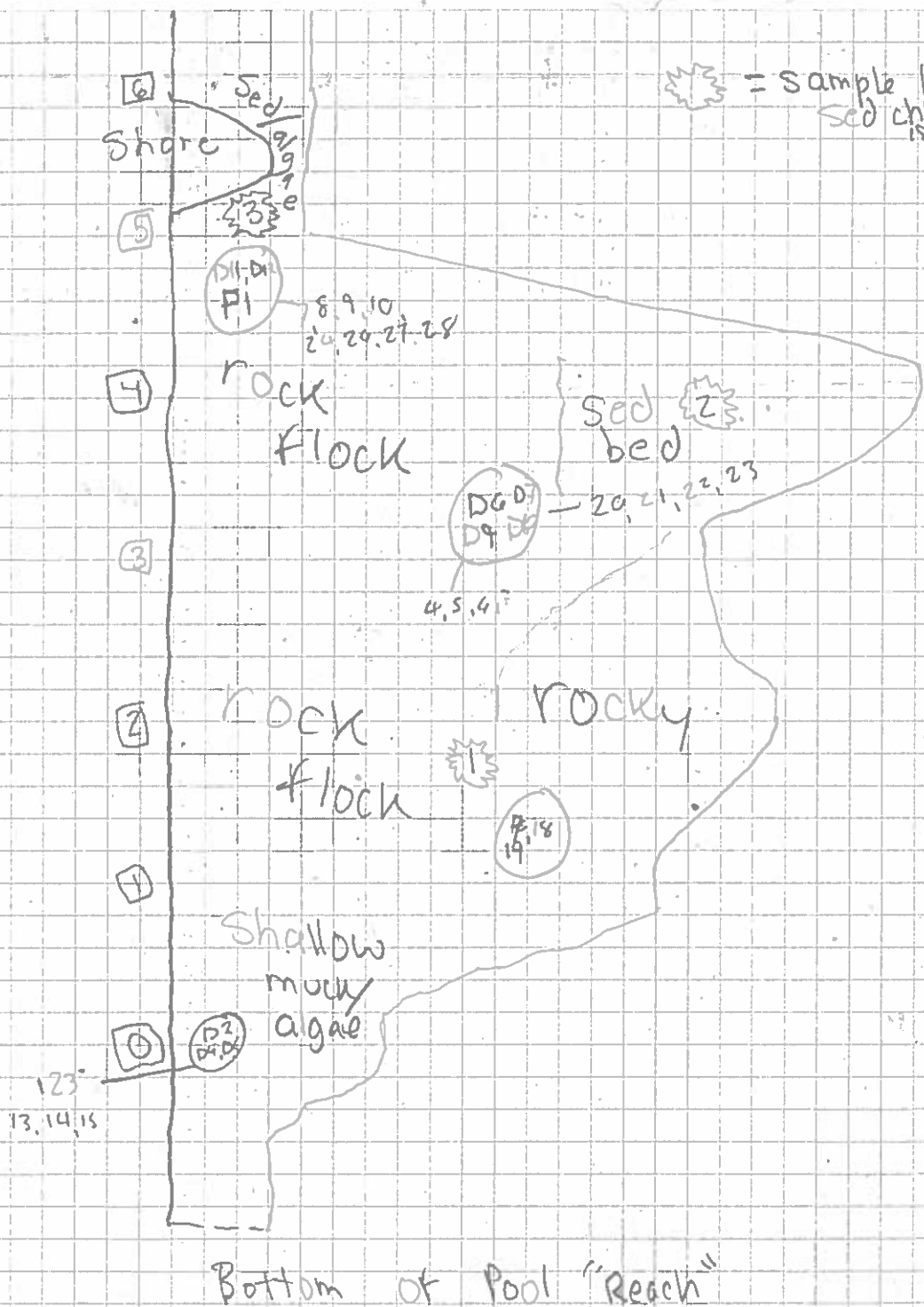
5/17/2018

Gridzzly.com - Make your own grid paper

□ = 1m vertical

□ = 0.25m horizontal

☼ = sample location
Sed chla isotopes, C







Hand-drawn map

$\frac{1}{\sqrt{\pi}} \int_0^x e^{-t} dt$



50

Q

Wings over wings ... 19 ... 11



Kings Creek High - KS 6-13-18 pg 4 of 13



Wings Creek High-KS 6-15-18 pg 2 of 1)



Cl 1000 6d

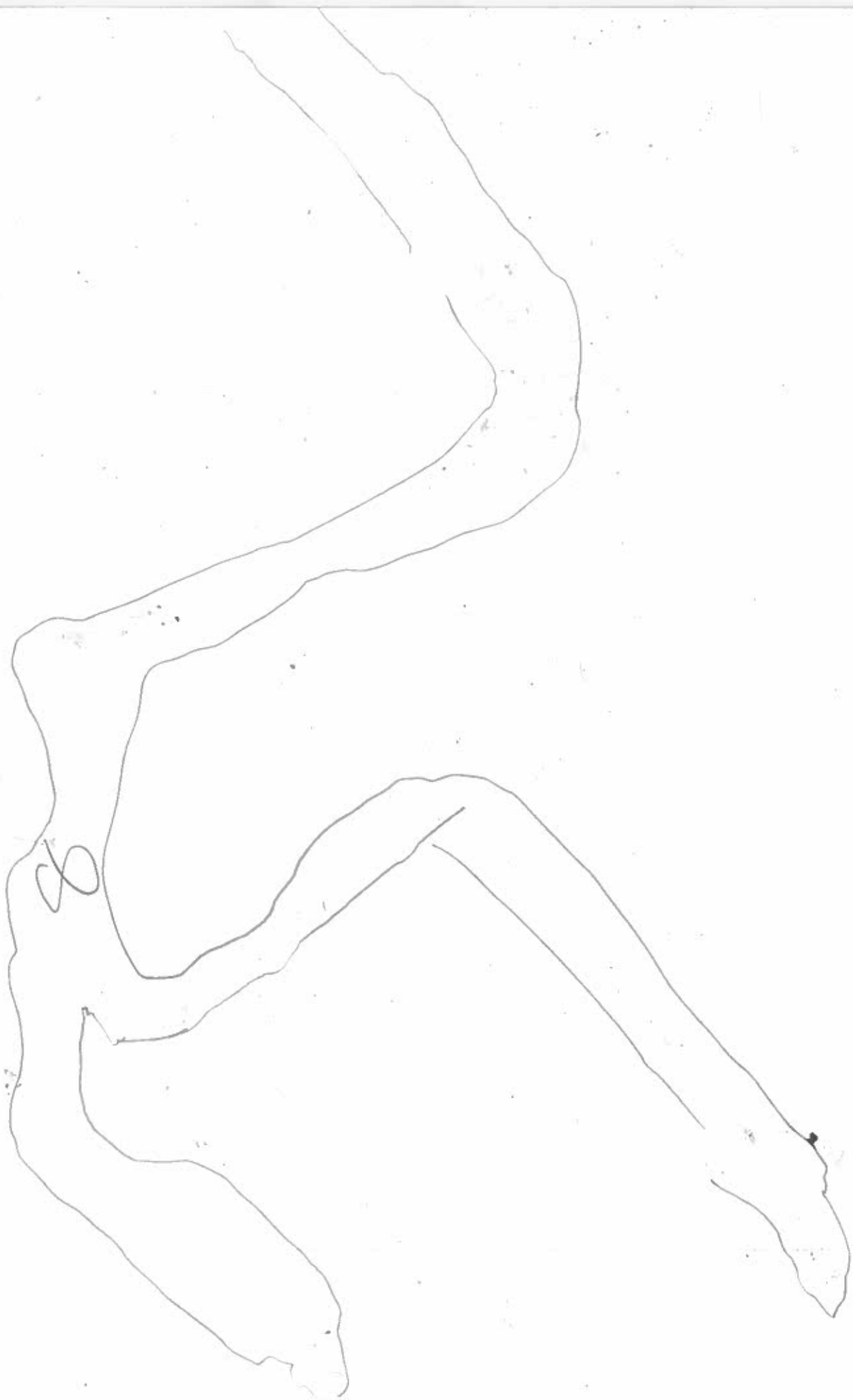
01-01-00

ca

no

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no



Kungs
Creek High

6-13-18

pg

7206-13



Kings Creek High - NS 6-13-18 pg 8 of 13

D 8

D 11

D 2

D 2

D 11

D 11

Kings creek high 6-13-18 pg 9 of 13

D2

D2

D1

D1D

D1

D1

king's ween high

b-13-18

pg 10 or 15

D/D

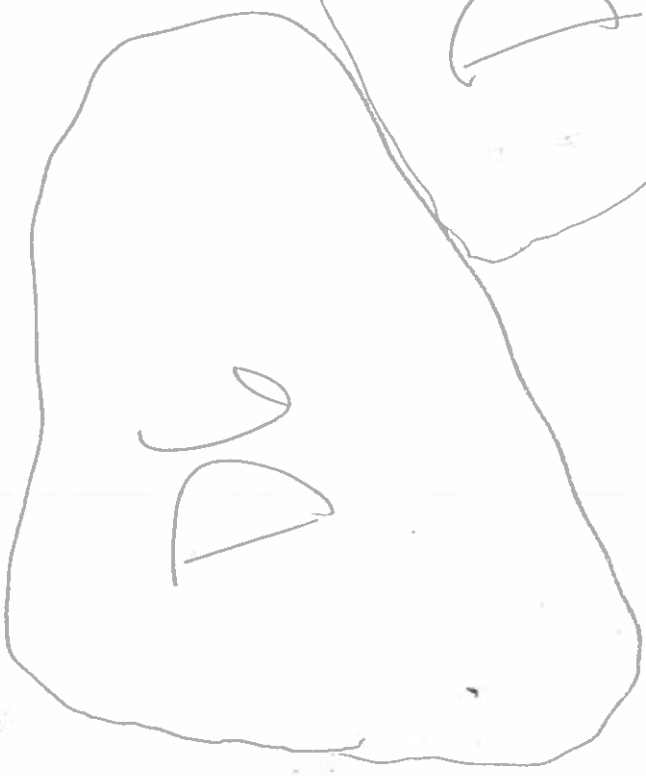
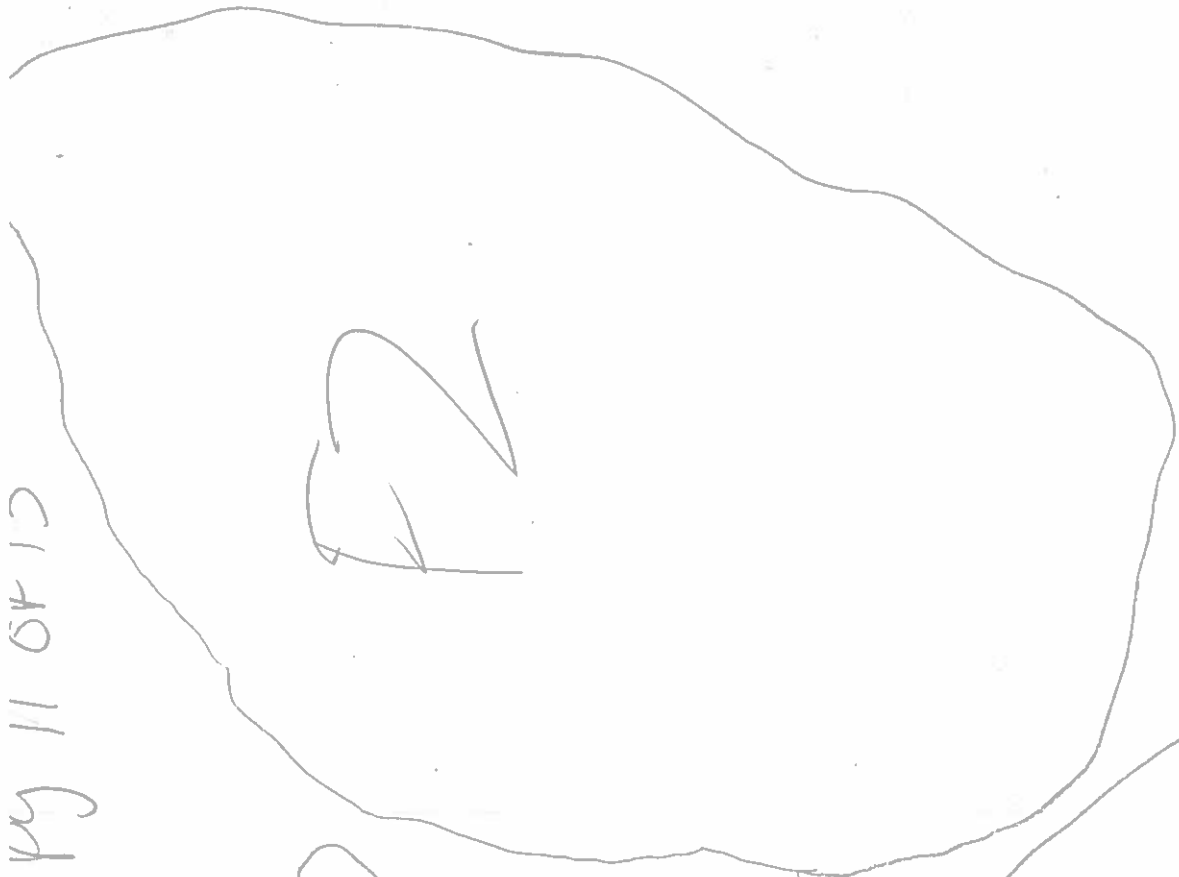
D/D

D 9

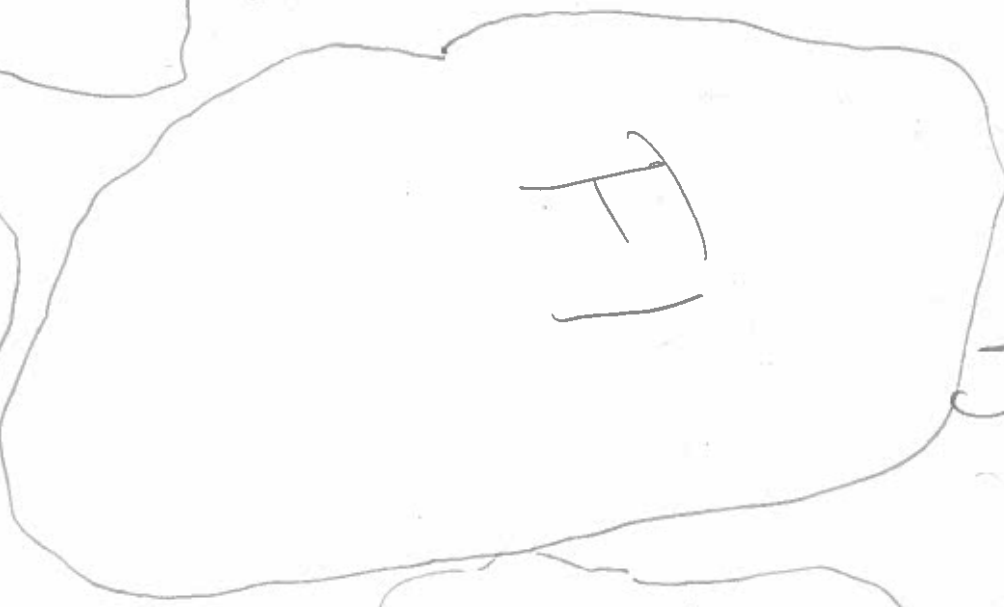
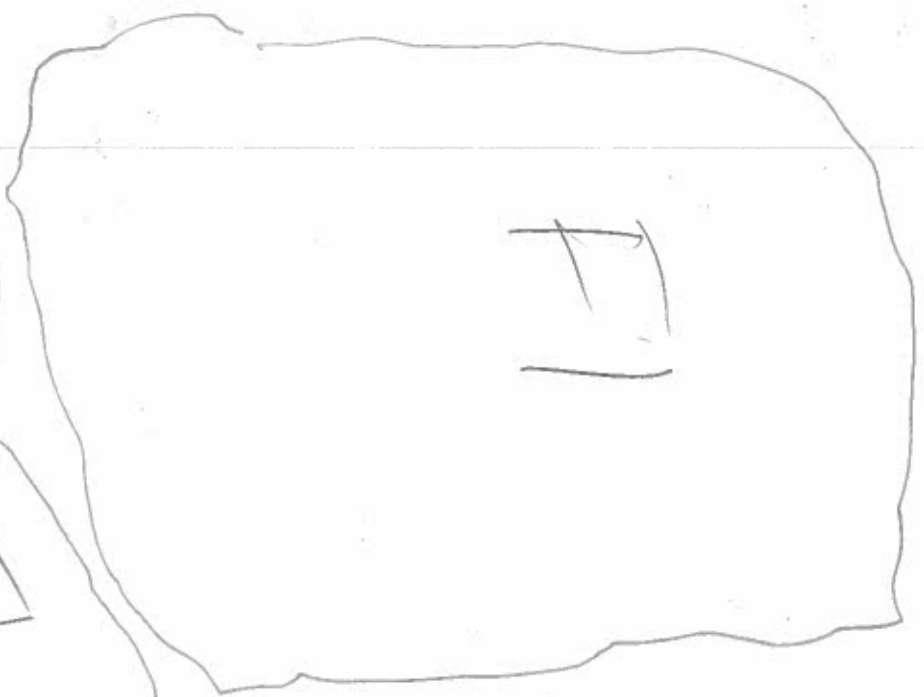
D 9

D 6

CI 10 11 6d
01-01-0
US 111
man shun



Kings were high, & 10-10 pg 10 0111



A hand-drawn diagram of a cell. The cell is represented by a large, irregular, rounded shape. Inside the cell, there is a smaller, roughly circular nucleus. Within the nucleus, there is a small, dense, dark-shaded area representing the nucleolus. A single line extends from the right side of the cell, representing a flagellum or tail.